



UNIVERSITÀ
DI TRENTO



Master's Degree in Cognitive Science

NEURAL MASS AND SPIKING MODELS FOR THE STUDY
OF NEURAL CIRCUITS

Supervisor:
Albrecht Haase

Student:
Zeno Rossi

Academic Year 2025/2026

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Chapter 1

Stochastic Resonance in a Spiking Neural Network Model of the Honeybee Antennal Lobe

1.1 Introduction

One of the first well established findings in neuroscience with the advent of electrical recordings of neuronal activity, was the variability of neurons' responses. That is, when stimulated with the same input, neurons respond with different temporal patterns of spikes, while the overall firing rate remains roughly equal [1], [2], [3].

This trial-to-trial variability is commonly called noise. Sources of noise in neurons are many and not all variability in response is due to noise (see [4] for a review). Still, due to the physical scale of cells, a stochastic component in neuronal function is inevitable by thermal noise alone. It is then to be expected that neural systems would have evolved to tolerate or, more daringly, exploit the presence of noise.

The idea of positive effects of noise on the dynamics of a system first appeared in the work of Benzi et al. [5] and Benzi et al. [6], where the term Stochastic Resonance (SR) was first used to describe the

Bibliography

- [1] G. J. Tomko and D. R. Crapper, “Neuronal variability: Non-stationary responses to identical visual stimuli,” *Brain Research*, vol. 79, no. 3, pp. 405–418, Oct. 25, 1974, issn: 0006-8993. DOI: 10.1016/0006-8993(74)90438-7. Accessed: Jun. 14, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0006899374904387>.
- [2] Z. F. Mainen and T. J. Sejnowski, “Reliability of Spike Timing in Neocortical Neurons,” *Science*, vol. 268, no. 5216, pp. 1503–1506, Jun. 9, 1995, issn: 0036-8075, 1095-9203. DOI: 10.1126/science.7770778. Accessed: Jun. 14, 2026. [Online]. Available: <https://www.science.org/doi/10.1126/science.7770778>.
- [3] D. J. Tolhurst, J. A. Movshon, and A. F. Dean, “The statistical reliability of signals in single neurons in cat and monkey visual cortex,” *Vision Research*, vol. 23, no. 8, pp. 775–785, Jan. 1, 1983, issn: 0042-6989. DOI: 10.1016/0042-6989(83)90200-6. Accessed: Jun. 14, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0042698983902006>.
- [4] A. A. Faisal, L. P. J. Selen, and D. M. Wolpert, “Noise in the nervous system,” *Nature Reviews Neuroscience*, vol. 9, no. 4, pp. 292–303, Apr. 2008, issn: 1471-0048. DOI: 10.1038/nrn2258. Accessed: Nov. 27, 2025. [Online]. Available: <https://www.nature.com/articles/nrn2258>.
- [5] R. Benzi, A. Sutera, and A. Vulpiani, “The mechanism of stochastic resonance,” *Journal of Physics A: Mathematical and General*, vol. 14, no. 11, p. L453, Nov. 1981, issn: 0305-4470. DOI: 10.1088/0305-4470/14/11/006. Accessed: Mar. 29, 2025. [Online]. Available: <https://dx.doi.org/10.1088/0305-4470/14/11/006>.
- [6] R. Benzi, G. Parisi, A. Sutera, and A. Vulpiani, “Stochastic resonance in climatic change,” *Tellus*, vol. 34, no. 1, pp. 10–16, 1982, issn: 2153-3490. DOI: 10.1111/j.2153-3490.1982.tb01787.x. Accessed: Jun. 16, 2026. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.2153-3490.1982.tb01787.x>.